Some solutions to Petersen's Riemannian geometry books

Jack Ceroni*

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I. Introduction

Current tally of Petersen problems solved:

- Chapter 1: 2/29
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II. Petersen solutions

A. Chapter 1

Solution II.1 (Exercise 1.6.3)**.** I'll do this problem for the case that M and N are compact oriented and p is orientation-preserving. Let $p : (M, g_M) \to (N, g_N)$ be the k-sheeted Riemannian cover. If U is an evenly-covered neighbourhood in N, then $p^{-1}(U) = U_1 \sqcup \cdots \sqcup U_k$, with each U_j a diffeomorphic copy of U. Of course, $g_M = p^* g_N$, so

$$\mathrm{vol}_M(X_1, \ldots, X_n) = \det[g_M(X_i, E_j)]_{i,j} = \det[g_N(p_* X_i, p_* E_j)]_{ij} = (p^* \mathrm{vol}_N)(X_1, \ldots, X_n) \tag{1}$$

which means, since p restricts to orientation-preserving diffeomorphisms $p : U_j \to U$, we have

$$\int_{U_j} (f \circ p) \ \mathrm{vol}_M = \int_{U_j} (f \circ p) \ p^* \mathrm{vol}_N = \int_U f \ \mathrm{vol}_N \tag{2}$$

for f a smooth function on U_j. In particular, if f is a smooth bump function supported in U, then each $(f \circ p)|_{U_1}, \ldots, (f \circ p)_{U_k}$ is a smooth bump function supported on $U_1, \ldots, U_k$. Thus, taking a partition of unity for the base and lifting to a partition of unity for the cover immediately yields:

$$\mathrm{vol}(M) = \sum_\alpha \sum_{j=1}^k \int_{U_{\alpha,j}} (f_\alpha \circ p)|_{U_{\alpha,j}} \ \mathrm{vol}_M = k \sum_\alpha \int_{U_\alpha} f_\alpha \ \mathrm{vol}_N = k \mathrm{vol}(N) \tag{3}$$

as desired.

* jceroni@uchicago.edu

Solution II.2 (Exercise 1.6.6)**.** Given coordinate chart $(U, \varphi) = (U, x^1, \ldots, x^n)$, we can write g as $g_{ij} dx^i \otimes dx^j$ inside U. Since the matrix $A = g_{ij}$ at a point in U is positive invertible, then $A^{-1/2}$ exists and $E_i = \sum_j (A^{-1/2})_{ij} \frac{d}{dx^j}$ form an orthonormal basis, as

$$g(E_i, E_j) = \sum_k (A^{-1})_{ik} A_{kj} = \delta_{ij} \tag{4}$$

Thus,

$$\mathrm{vol}\left(\frac{d}{dx^1}, \ldots, \frac{d}{dx^n}\right) = \det\left[g\left(\frac{d}{dx^i}, E_j\right)\right]_{i,j} = \det\left[\sum_k A_{ik}(A^{-1/2})_{kj}\right]_{ij} = \det\sqrt{A} \tag{5}$$

Of course, the $E_1, \ldots, E_n$ can be made oriented by flipping the sign of E_1 if necessary, which has the effect of flipping the sign of the determinant. In particular, the $E_1, \ldots, E_n$ frame will be oriented if the coordinates $(x^1, \ldots, x^n)$ are an oriented chart.

B. Chapter 2

Solution II.3 (Exercise 2.5.5)**.** To begin, note that

$$\mathrm{div}(fX)\mathrm{vol} = d\iota_{fX}\mathrm{vol} = d(f\iota_X\mathrm{vol}) = df \wedge \iota_X\mathrm{vol} + f\mathrm{div}(X)\mathrm{vol} \tag{6}$$

Note that $df \wedge \mathrm{vol} = 0$, hence $0 = \iota_X df \wedge \mathrm{vol} - df \wedge \iota_X\mathrm{vol}$, so we obtain $\mathrm{div}(fX) = X(f) + f\mathrm{div}(X)$, as desired. Next, note that

$$\begin{aligned}
\Delta(fh) = \mathrm{div}(\nabla(fh)) = \mathrm{div}(d(fh)^\#) = \mathrm{div}(hdf^\# + fdh^\#) &= \mathrm{div}(h\nabla f) + \mathrm{div}(f\nabla h) && (7)\\
&= \nabla f(h) + \nabla h(f) + f\Delta h + h\Delta f && (8)\\
&= dh(\nabla f) + df(\nabla h) + f\Delta h + h\Delta f && (9)\\
&= 2g(\nabla f, \nabla h) + f\Delta h + h\Delta f && (10)
\end{aligned}$$

as desired. Finally,

$$\mathrm{Hess}(fh) = \frac{1}{2}\mathcal{L}_{\nabla(fh)}g = \frac{1}{2}\mathcal{L}_{f\nabla h}g + \frac{1}{2}\mathcal{L}_{h\nabla f}g \tag{11}$$

Recall that

$$(\mathcal{L}_{f\nabla h}g)(X, Y) = f(\mathcal{L}_{\nabla h}g)(X, Y) + X(f)g(\nabla h, Y) + Y(f)g(X, \nabla h) \tag{12}$$

so immediately,

$$\begin{aligned}
\mathrm{Hess}(fh)(X, Y) &= \frac{1}{2}\Big(X(f)g(\nabla h, Y) + Y(f)g(X, \nabla h) + X(h)g(\nabla f, Y) + Y(h)g(X, \nabla f)\Big) && (13)\\
&\quad + f\mathrm{Hess}(h)(X, Y) + h\mathrm{Hess}(f)(X, Y) && (14)\\
&= (df \otimes dh)(X, Y) + (dh \otimes df)(X, Y) + f\mathrm{Hess}(h)(X, Y) + h\mathrm{Hess}(f)(X, Y) && (15)
\end{aligned}$$

which is what we wanted.

Solution II.4 (Exercise 2.5.7)**.** We have the invariant formula $d\theta_X(Y, Z) + (\mathcal{L}_X g)(Y, Z) = 2g(\nabla_Y X, Z)$. Moreover,

$$X(g(Y, Z)) = (\mathcal{L}_X g)(Y, Z) + g([X, Y], Z) + g(Y, [X, Z]) \tag{16}$$

Moreover, from the properties of the connection,

$$\begin{aligned}
X(g(Y, Z)) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z) && (17)\\
\nabla_X Y - \nabla_Y X - [X, Y] &= \nabla_X Z - \nabla_Z X - [X, Z] = 0 && (18)
\end{aligned}$$

so when we substitute the commutators, and cancel, we obtain $(\mathcal{L}_X g)(Y, Z) = g(\nabla_Y X, Z) + g(Y, \nabla_Z X)$, which then gives the desired result.

Solution II.5 (Exercise 2.5.11)**.** The condition that the map $Z \mapsto \nabla_Z X$ is self-adjoint is simply

$$g(\nabla_Y X, Z) - g(Y, \nabla_Z X) = 0 \quad \text{for all } Y, Z \tag{19}$$

From Exercise 2.5.7, we know that this is equivalent to $d\theta_X(Y, Z) = 0$ for all Y and Z, which means that θ_X is a closed 1-form, which is equivalent to being locally exact. In particular, $\theta_X = df$ locally and $X = (df)^\#$, which completes the proof.

Solution II.6 (Exercise 2.5.12)**.** There are two parts:

1. This follows immediately from Koszul's formula and naturality of the Lie bracket.

2. Clearly, any F of the form $F(x) = Ox + b$ is an isometry. Conversely, if F is an isometry, then note that $\nabla_{\partial_i} F_* \partial_j = F_*(\nabla_{(F^{-1})_* \partial_i} \partial_j) = 0$, hence $F_* \partial_j$ is a constant vector field $O^{jk} \partial_k$ for constants O^{jk}. Since we are working in Euclidean space, $DF = O = [O^{jk}]_{j,k}$ everywhere, which means that $F(x) = Ox + b$. It is obvious that O is an isometry, as

$$\delta_{ij} = g(\partial_i, \partial_j) = g(F_* \partial_i, F_* \partial_j) \tag{20}$$

everywhere.

This next exercise is important, it demonstrates the existence of normal frames.

Solution II.7 (Exercise 2.5.19)**.** To begin, choose some arbitrary orthonormal frame near p, and then multiply by some constant element of $\mathrm{O}(n)$ so that our new frame $\overline{E}_i$ satisfies $\overline{E}_i(p) = e_i$ for each i. From here, we wish to define $E_i = \alpha_i^j \overline{E}_j$ such that

$$0 = \nabla_{e_k} E_i = e_k(\alpha_i^j)\overline{E}_j + \alpha_i^j \nabla_{e_k} \overline{E}_j = e_k(\alpha_i^\ell)\overline{E}_\ell + \alpha_i^j(p)\Gamma_{kj}^\ell \overline{E}_\ell = e_k(\alpha_i^\ell)\overline{E}_\ell + \Gamma_{ki}^\ell \overline{E}_\ell \tag{21}$$

so we simply require that each of the matrix elements α_i^ℓ have particular directional derivatives at p. Note that $\Gamma_{ki}^\ell = -\Gamma_{k\ell}^i$ from naturality of the connection, hence each matrix Γ_k is skew-Hermitian. Note that $e_k = e_k^i \partial_i$ for some coordinates $(U, x^1, \ldots, x^n) = (U, \varphi)$ centred around p. Let $y^k = \sum_{ij} g_{ij} e_k^i x^j$ with $g_{ij} = g_p(\partial_i, \partial_j)$. We thus define

$$A(q) = \exp\left(-\sum_k y^k(q)\Gamma_k\right) \tag{22}$$

and note that $A(p) = \mathrm{id}$. In coordinates,

$$A \circ \varphi^{-1}(s^1, \ldots, s^n) = \exp\left(-\sum_{ijk} g_{ij} e_k^i s^j \Gamma_k\right) \tag{23}$$

and the corresponding derivatives at the origin are clearly given by

$$\sum_j e_\ell^j \frac{d}{ds^j}(A \circ \varphi^{-1})(0) = -\sum_{ijk} g_{ij} e_\ell^j e_k^i \Gamma_k = -\sum_k \delta_{\ell k}\Gamma_k = -\Gamma_\ell \tag{24}$$

which means, immediately, that if we let $\alpha_i^\ell(q) = A(q)_{i\ell}$ then $e_k(\alpha_i^\ell)|_p = -\Gamma_{ki}^\ell$ as desired. Of course, since each Γ_k is skew-Hermitian, then $A(q) \in \mathrm{O}(n)$ for all q, so our new frame consisting of the E_j is still orthonormal.

This exercise is also important, it demonstrates the existence of normal coordinates:

Solution II.8 (Exercise 2.5.22)**.** Clearly, $\varphi^{-1} = (u^1, \ldots, u^{n+m})$ has well-defined local inverse on M, giving rise to a coordinate system $\varphi = (x^1, \ldots, x^n)$. If $\iota : M \to \mathbb{R}^{n+m}$ is the inclusion of the submanifold, then $y_j \circ \iota \circ \varphi^{-1} = u^j$, hence

$$\iota^*(dy^j) = d(y^j \circ \iota) = \sum_{i=1}^n \frac{d(y^j \circ \iota)}{dx^i} dx^i = \sum_{i=1}^n \frac{du^j}{ds^i} dx^i \tag{25}$$

which immediately implies the desired result for the metric. To obtain the desired formula for the Christoffel symbols, note that in some set of coordinates,

$$\Gamma_{ij,\ell} = \frac{1}{2}(\partial_j g_{i\ell} + \partial_i g_{j\ell} - \partial_\ell g_{ij}) \tag{26}$$

Clearly,

$$\frac{dg_{ij}}{ds^\ell} = \sum_{k=1}^{n+m} \frac{d^2u^k}{ds^\ell ds^i}\frac{du^k}{ds^j} + \frac{d^2u^k}{ds^\ell ds^j}\frac{du^k}{ds^i} \tag{27}$$

so when we plug this formula into the above equation and re-index, we obtain the desired result.

Solution II.9 (Exercise 2.5.26)**.** There are multiple parts:

1. Recall that $\nabla_X Y$ is well-defined when X is given at only a single point p, and when Y is specified along a curve γ such that $\dot\gamma(0) = X_p$. Clearly, in the case of the problem statement, X is tangent to M, so we can always find a curve γ in M such that $\dot\gamma(0) = X_p$. The vector field V is defined on all of M, hence on γ.

2. As is noted in the problem statement, we let

$$\nabla_X^{\overline{M}} V = \nabla_X^\perp V + T_X V \tag{28}$$

be the decomposition into normal and tangential components. Now, we note that for Y tangent to M,

$$g_{\overline{M}}(\nabla_X^{\overline{M}} Y - \nabla_X^M Y, V) = g_{\overline{M}}(\nabla_X^{\overline{M}} Y, V) = X(g_{\overline{M}}(Y,V)) - g_{\overline{M}}(Y, \nabla_X^{\overline{M}} V) \tag{29}$$

$$= -g_M(Y, T_X V) \tag{30}$$

as $g_{\overline{M}}(Y,V) = 0$ everywhere.

3. By definition, $\nabla_X^\perp V$ is the projection of $\nabla_X^{\overline{M}} V$ onto $T^\perp M$, which is itself a linear operation on each tangent space. The fact that $\nabla_X^{\overline{M}} V$ is linear in both variables, a derivation in V and tensorial in X immediately implies $\nabla_X^\perp V$ is.

Solution II.10 (Exercise 2.5.29)**.** There are multiple parts:

1. Suppose in a neighbourhood of point p we have $\mathrm{grad}(f) = X$, for some function $f : U \to \mathbb{R}$. Of course, $\langle Y, X\rangle = df(Y) = Y(f) = 0$ for any Y which is orthogonal to X (locally). However, $X(f) = g(X,X) = 1$, as X is a unit vector field, so f is a submersion, and $S = f^{-1}(0)$ is an $(n-1)$-dimensional submanifold of $U \subset M$, where $T_q S = \mathrm{Ker}(f_*) = \mathrm{Ker}(df)$, which is precisely the the orthogonal distribution. Conversely, note that showing X is locally a gradient field is equivalent to showing that $X^\flat$ is closed (equivalent to locally exact). Thus, we want to show that

$$dX^\flat(Y,Z) = Y(g(X,Z)) - Z(g(X,Y)) - g(X,[Y,Z]) \tag{31}$$

If Y, Z are both in the orthogonal distribution, then $[Y,Z]$ is, by integrability, so the final term vanishes. Moreover, the first two terms clearly vanish as well. If we set $Y = X$ with Z still in the orthogonal distribution, then the above reduces to

$$-g(X,[X,Z]) = g(X, \nabla_Z X) - g(X, \nabla_X Z) = \frac{1}{2}Z(g(X,X)) + g(\nabla_X X, Z) - X(g(X,Z)) = 0 \tag{32}$$

from the assumption that $\nabla_X X = 0$. It is also easy to see that $dX^\flat$ vanishes on $(Y,Z) = (X,X)$ and when $Y = X$ (by symmetry). Thus, $dX^\flat = 0$, so $X^\flat$ is closed, hence locally exact, so X is locally a gradient field.

2. This is just the definition of integrability?

3. TODO

Solution II.11 (Exercise 2.5.30)**.** There are multiple parts:

1. Note that if X, Y are tangent to E (or F), then

$$[X, Y] = \nabla_X Y - \nabla_Y X \tag{33}$$

is also tangent to E (or F), by the parallel assumption, so Frobenius' theorem implies that the distributions are integrable.

2. We can always assume that the integral submanifolds are open, hence $U = V_E \times V_F$ is an open set around a particular point. Moreover, since E and F are orthogonal complements of each other, TU splits as $TU = TV_E \times TV_F \simeq E|_{V_E} \oplus F|_{V_F}$, the Whitney sum, and any $v \in T_pM$ can be naturally split as $v = v_E \oplus v_F$ with $v_E \in T_pV_E$ and $v_F \in T_pV_F$ (pushed-forward via inclusion of these submanifolds into M). Then,

$$g_p(v, w) = g_p(v_E \oplus v_F, w_E \oplus w_F) = (g|_{V_E})_p(v_E, w_E) + (g|_{V_F})_p(v_F, w_F) \tag{34}$$

from the orthogonality assumption.

Solution II.12 (Exercise 2.5.31)**.** When we say X is parallel, we mean that $\nabla X = 0$. Thus, $Y(g(X, X)) = 2g(\nabla_Y X, X) = 0$ for any vector field Y, so $g(X, X)$ must be constant, so X is constant-length. Let $D = X^\perp$ be the orthogonal distribution. Then given Y, Z parallel to D, note that $g(\nabla_Y Z, X) = Y(g(Z, X)) - g(Z, \nabla_Y X) = 0$, so $\nabla_Y Z$ is parallel to D as well, hence D is a parallel distribution. It follows from the previous exercise (2.5.30) that locally, the metric is of the form $g|_V + dt^2$. In particular, the integral submanifold to X is a curve (if $X \neq 0$ everywhere), and this curve will be locally diffeomorphic to an interval.

C. Chapter 3

Solution II.13 (Exercise 3.4.1)**.** This more or less boils down to writing down R_{ijkl} using the result derived in Exercise 2.5.22, and then showing that we have cancellation of the terms involving third derivatives.

Solution II.14. Exercise 3.4.2] Note that if X is some vector field, then

$$\begin{aligned}
\frac{1}{2}X(g(\mathrm{grad}(f), \mathrm{grad}(f))) &= g(\nabla_X \mathrm{grad}(f), \mathrm{grad}(f)) && (35)\\
&= g(\nabla_{\mathrm{grad}(f)} X, \mathrm{grad}(f)) + g([X, \mathrm{grad}(f)], \mathrm{grad}(f)) && (36)\\
&= g(\nabla_{\mathrm{grad}(f)} X, \mathrm{grad}(f)) + df([X, \mathrm{grad}(f)]) && (37)\\
&= g(\nabla_{\mathrm{grad}(f)} X, \mathrm{grad}(f)) + X(g(\mathrm{grad}(f), \mathrm{grad}(f))) - \mathrm{grad}(f)(g(X, \mathrm{grad}(f))) && (38)\\
&= g(X, \nabla_{\mathrm{grad}(f)} \mathrm{grad}(f)) + X(g(\mathrm{grad}(f), \mathrm{grad}(f))) && (39)
\end{aligned}$$

which implies that $-2g(X, \nabla_{\mathrm{grad}(f)} \mathrm{grad}(f)) = X(|\mathrm{grad}(f)|^2)$. It follows immediately that $|\mathrm{grad}(f)|$ is constant if and only if $\nabla_{\mathrm{grad}(f)} \mathrm{grad}(f) = 0$. Obviously this means that (2) implies (3) as well. However, (3) does not imply (2): consider $f(x, y) = x^2 + y^2$ on $\mathbb{R}^2$. Obviously, $\mathrm{grad}(f) = 2(x, y)$ so $|\mathrm{grad}|^2 = 4f(x, y)$, which is constant on level-sets, but $\nabla_{\mathrm{grad}(f)} \mathrm{grad}(f) = 2\mathrm{grad}(f)$.

Solution II.15 (Exercise 3.4.3)**.** We know that

$$(\nabla_{\nabla f} S)(X, \omega) + \omega(S(\nabla_{\nabla f} X)) + (\nabla_{\nabla f}\omega)(S(X)) = \nabla f(S(X, \omega)) = (L_{\nabla f} S)(X, \omega) + \omega(S(L_{\nabla f} X)) + (L_{\nabla f}\omega)(S(X)) \tag{40}$$

Clearly,

$$(\nabla_{\nabla f}\omega)(S(X)) = (\nabla f)(\omega(S(X))) - \omega(\nabla_{\nabla f}S(X)) \tag{41}$$

where

$$\nabla_{\nabla f}S(X) = [\nabla f, S(X)] + \nabla_{S(X)}\nabla f = [\nabla f, S(X)] + S(S(X)) \tag{42}$$

as well as

$$(L_{\nabla f}\omega)(S(X)) = (\nabla f)(\omega(S(X))) - \omega([\nabla f, S(X)]) \tag{43}$$

Therefore, after some cancellation, we get

$$(\nabla_{\nabla f}S)(X,\omega) + \omega(S(\nabla_{\nabla f}X)) - \omega(S(S(X))) = (L_{\nabla f}S)(X,\omega) + \omega(S([\nabla f, X])) \tag{44}$$

and finally, note that substituting

$$\nabla_{\nabla f}X = \nabla_X \nabla f + [\nabla f, X] = S(X) + [\nabla f, X] \tag{45}$$

gives the desired result. As for the second formula, I don't want to check it right now.

Solution II.16 (Exercise 3.4.4)**.** Under the assumption that X, Y, Z are tangent to H, note $g(S(X), Y) = \Pi(X, Y)$ (as $N = \nabla r$ in this context), then note

$$g((S(X)\wedge S(Y))Z, W) = g(S(X), Z)g(S(Y), W) - g(S(Y), Z)g(S(X), W) = \Pi(X, Z)\Pi(Y, W) - \Pi(Y, Z)\Pi(X, W) \tag{46}$$

which immediately gives the first formula. The other two formulas follow similarly.

Solution II.17 (Exercise 3.4.6)**.** Suppose M has constant curvature k, then we are able to make use of the formula of Proposition 3.1.3, which tells us that

$$R(X, Y)Z = -k\left(g(X, Z)Y - g(Y, Z)X\right) \tag{47}$$

which means that

$$R(X, Y, Z, W) = -k\left(g(X, Z)g(Y, W) - g(Y, Z)g(X, W)\right) \tag{48}$$

From here, it is very easy to verify that R is parallel, using compatibilitiy of the Levi-Civita connection with the metric.

Solution II.18 (Exercise 3.4.7)**.** Note that

$$0 = (\nabla_Z \mathrm{Ric})(X, Y) = Z(\mathrm{Ric}(X, Y)) - \mathrm{Ric}(\nabla_Z X, Y) - \mathrm{Ric}(X, \nabla_Z Y) \tag{49}$$

so using symmetry of the Ricci tensor, $Z(\mathrm{Ric}(X, X)) = 2\mathrm{Ric}(\nabla_Z X, X)$. From here, we can choose an orthonormal local frame around some p, denoted $E_1, \ldots, E_n$ such that $\nabla_v E_j = 0$ for all $v \in T_pM$. In particular, $Z(\mathrm{Ric}(E_j, E_j))(p) = 0$ for all j, so $Z(\mathrm{scal})(p) = 0$. This works for any p, so scal is constant.

Solution II.19 (Exercise 3.4.9)**.** Obviously this follows from writing X and Y in local coordinates for $M \times N$, and noting that $\nabla_X Y = 0$ immediately. It then is immediately obvious that $R(X, Y) = 0$ as well, so $\sec(X, Y) = 0$.

Solution II.20 (Exercise 3.4.10)**.** One implication is easy: if M has constant sectional curvature at p then $R(v, w)z = -k(g(v, z)w - g(w, z)v) = 0$ for all orthogonal $v, w, z \in T_pM$. Conversely, suppose $R(v, w)z = 0$ for all such tangent vectors. Recall that we have the curvature bilinear form $R : \wedge^2 TM \times \wedge^2 TM \to \mathbb{R}$ given

by $R(x \wedge y, w \wedge z) = R(x, y, z, w) = g(R(x, y)z, w)$ (extended bilinearly). If we pick an orthonormal basis $e_1, \ldots, e_n$ for T_pM, then the collection of $e_i \wedge e_j$ with $i < j$ will form an orthonormal basis for $\wedge^2 T_pM$, as

$$g(e_i \wedge e_j, e_k \wedge e_\ell) = g(e_i, e_k)g(e_j, e_\ell) - g(e_i, e_\ell)g(e_j, e_k) = \delta_{ik}\delta_{j\ell} - \delta_{i\ell}\delta_{jk} \tag{50}$$

which is 1 if $i = k$ and $j = \ell$, -1 if $i = \ell$ and $j = k$, and 0 otherwise. But the latter case can actually never happen, as we would have $j < i$ as $k < \ell$, which is a contradiction. From here, note that

$$R(e_i \wedge e_j, e_k \wedge e_\ell) = g(R(e_i, e_j)e_\ell, e_k) \tag{51}$$

We know that when $\ell \neq i, j$ then $R(e_i, e_j)e_\ell = 0$. Similarly, when $k \neq i, j$, we can use symmetry to conclude that the above vanishes as well. Finally, when $k = \ell$, the corresponding term vanishes. Thus, we have

$$R(e_i \wedge e_j, e_i \wedge e_j) = \lambda^{ij} = R(e_j \wedge e_i, e_j \wedge e_i) = \lambda^{ji} \tag{52}$$

for some constants λ^{ij} ranging over *all* i and j, and $R = 0$ otherwise. Clearly $\lambda^{jj} = 0$. From here, pick $e_i \wedge e_j$ and ℓ with $i, j \neq \ell$. Then

$$R(e_i \wedge (e_j + e_\ell), e_i \wedge (e_j - e_\ell)) = g(R(e_i, e_j + e_\ell)(e_j - e_\ell), e_i) = 0 \tag{53}$$

On the other hand, the above is equal to $\lambda^{ij} - \lambda^{i\ell}$. It therefore follows that

$$\lambda^{ij} = \lambda^{i\ell} = \lambda^{\ell i} = \lambda^{\ell k} = \lambda^{k\ell} \tag{54}$$

for any pairs $i < j$ and $k < \ell$. Thus, R is a multiple of the metric on a fixed orthonormal basis, and we therefore have constant curvature.

Solution II.21 (Exercise 3.4.12)**.** Of course, the Ricci curvature vanishes identically in $\mathbb{R}^n$. Therefore, the tangential curvature equation takes the form

$$g_H(R^H(X, Y)Z, W) = \Pi(X, W)\Pi(Y, Z) - \Pi(X, Z)\Pi(Y, W) \tag{55}$$

which tells us that

$$\mathrm{Ric}^H(X, Y) = \sum_{i=1}^{n-1} g_H(R^H(E_i, X)Y, E_i) = \sum_{i=1}^{n-1} \Pi(E_i, E_i)\Pi(X, Y) - \Pi(X, E_i)\Pi(E_i, Y) \tag{56}$$

where $E_1, \ldots, E_{n-1}$ is a local orthonormal frame for TH^{n-1}. The $(1, 1)$ tensor S obtained from Π satisfies $\Pi(X, Y) = g(S(X), Y) = g(X, S(Y))$, hence

$$\mathrm{Ric}^H(X, Y) = \sum_i g(S(E_i), E_i)g(S(X), Y) - g\left(S(X), \sum_i g(E_i, S(Y))E_i\right) = \mathrm{Tr}(S)g(S(X), Y) - g(S^2(X), Y) \tag{57}$$

Hence, as a $(1, 1)$ tensor, Ric^H is $\mathrm{Tr}(S)S - S^2$.

Solution II.22 (Exercise 3.4.13)**.**

1. Note that S^n_k is isotropic, so without loss of generality, we may assume that the tangent vector in question is $v = (0, \ldots, 1)$ at the point $p = (1, 0, \ldots, 0)$. Consider the embedded copy of S^{n-1}_k (denoted H) consisting of all points $(x^0, \ldots, x^{n-1}, 0)$. We can extend V to a normal vector field on H in the obvious way, and note that clearly, the second fundamental form for H vanishes (in fact V is a parallel vector field).

2. Given $v, w, z \in T_pM$, if the vectors are mutually orthogonal, then by the assumption, v is normal to a totally geodesic hypersurface H, and as a result, we must have $w, z \in T_pH$. From here, we already know that $g(R(w, z)v, v) = 0$. If we pick $y \in T_pH$, then $g(R(w, z)v, y) = -g(R(w, z)y, v)$, and we are able to compute this quantity via the mixed curvature equation. But because $\Pi^H = 0$, this implies that $g(R(w, z)v, y) = 0$. Thus, $R(w, z)v = 0$. This holds for any choice of orthogonal v, w, z, so it follows from Exercise 3.4.10 that M has constant curvature.